

TOPIC 10. COMPOSITE MATERIALS FOR AEROSPACE INDUSTRY. (Session II)

Why polymers are used in Composite Materials?

- Polymer matrices in Composite Materials

- Thermosetting matrices

- Thermoplastic matrices

- The interface region

Mean elastic properties of composites

- Mechanics of materials approach

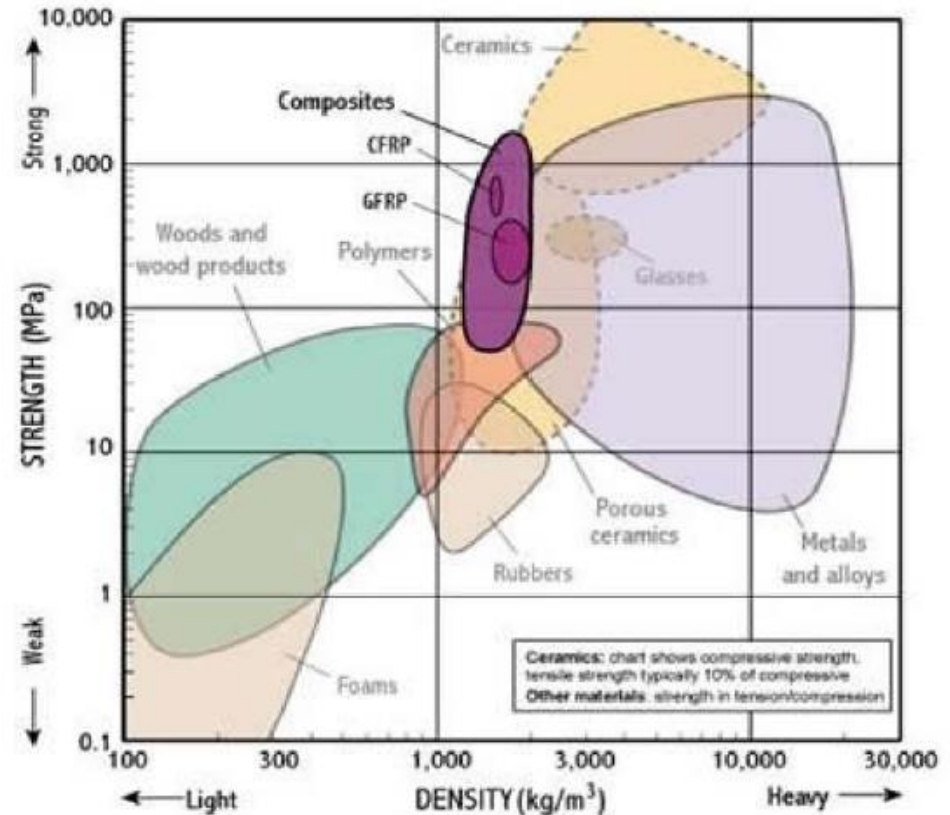
- Elastic deformation of long-fibre composites

- Elastic deformation in composite with an uniformly dispersed aggregate

Why polymers are used in Composite Materials?

Advantages of advanced polymer composite materials

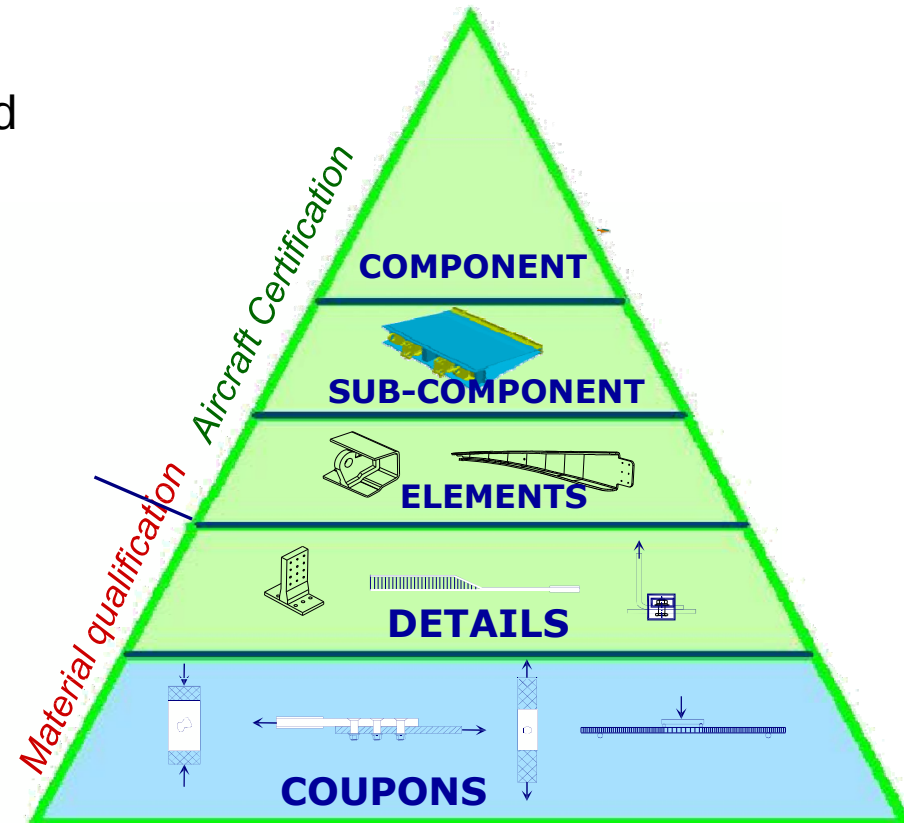
- Good specific mechanical performance
- Good fatigue performance
- Low weight
- Custom design
- Dimensional stability
- High chemical resistance
- No corrosion problems
- Good fire behavior
- Reduction in number of parts



Why polymers are used in Composite Materials?

Disadvantages of advanced polymer composite materials

- Costly programs for evaluation, and certification of structures
- Costly investments for installations and equipment
- High material cost
- Water entry in sandwich structures
- Low impact resistance
- Additional protection for erosion
- Low thermal conductivity. Additional protection for lightning
- Low recyclability



POLYMER MATRICES: THERMOSET AND THERMOPLASTIC**PROPERTIES****Thermosetting polymers**

- **Croslinked chains**
- **Insoluble (swell)**
- **Amorphous (T_g)**
- **Low shelf life**
- **Do not flow under stress even at high T.**
- **Brittle (low strains at failure)**
- **Sensitivity to moisture**

Thermoplastic matrices

- **Semicrystalline. High T_m**
- **Consolidation: (No chemical reaction) processing by heat and pressure**
- **Potentially reprocessed**
- **Large shelf life**
- **Potential for fast part fabrication**
- **Very high melt viscosities (high pressure and high shearing forces during manufacturing → Reinforcement damage)**

POLYMER MATRICES: THERMOSET AND THERMOPLASTIC

THERMOSET

THERMOPLASTIC

Main Characteristics

- Undergoes chemical change when cured
- low strain to failure
- low fracture energy
- processing is irreversible
- very low viscosity possible
- absorbs moisture
- highly resistant to solvents

- Non-reacting, no curing required
- high strain to failure
- high fracture energy
- processing is reversible
- very high viscosity
- absorbs little moisture
- limited resistance to organic solvents, in some case

Advantages

- Relatively low processing temperature
- good fiber wetting
- formable into complex shapes
- liquid-resin manufacturing feasible
- resistant to creep

- Short processing times possible
- reusable scrap
- post-formable can be reprocessed
- rapid processing
- unlimited shelf life without refrigeration
- high delamination resistance

Disadvantages

- long processing time and long cure (1-2 h)
- restricted storage life

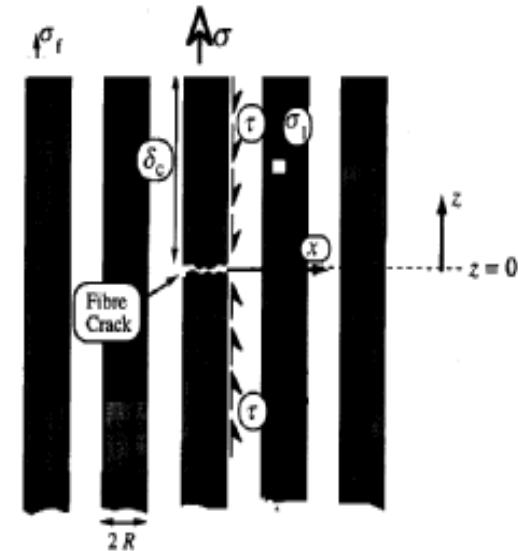
- lower resistance to solvents
- requires high T (300-400 °C) and P processing
- can be prone to creep

POLYMER MATRICES: THERMOSET AND THERMOPLASTIC

FIBER-REINFORCED COMPOSITE MATERIALS

POLYMER MATRIX FUNCTIONS

- Transfer load from into and out of the fibers
- Protects the fiber from the environment (chemical degradation)
- Separates the fibres to prevent failure of adjacent fibers if one fails (Avoid crack propagation)
- Maintain the orientation of the fibers
- Form the shape of the component

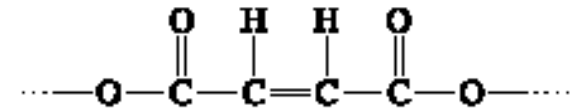


Thermosetting Matrix Materials

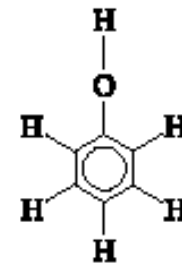
Thermosetting matrices used in the aerospace industry

- Epoxy
 - With elastomeric additives
 - With thermoplastic additive
- BMI
- PI (thermoplastic or thermosetting)
- Polyester
- Phenolic

Polyesters



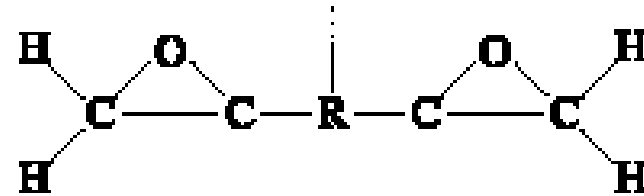
Phenolics



Thermosetting Matrix Materials

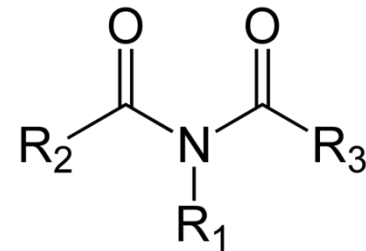
EPOXY

- They are the most used in structural applications.
- Good mechanical properties in general, the temperature can vary between 60-150 ° C.
- They have low shrinkage and good adhesion to most fibers.
- Usually cured at 120 ° C and at 180 ° C).



Polyimides (PI)

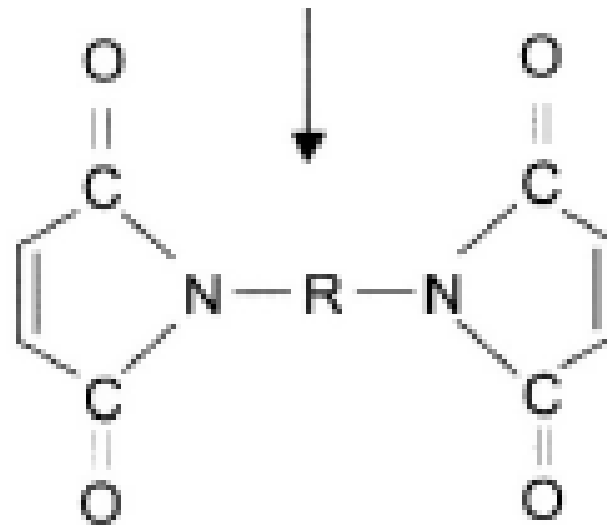
- More expensive and less used than epoxy resins.
- Designed to work with relatively high service temperatures between 130 and 230 ° C
- In general, all of these resins have a higher water absorption and lower stiffness of epoxy resins



Thermosetting Matrix Materials

Bismaleimides (BMI)

- Tg between 275 and 300 ° C. Good performance at high service temperatures (250 ° C)
- Can use conventional curing cycles, yielding laminates with low porosity content.
- Usually a need for post-curing to achieve optimum properties
- Good mechanical properties.
- Present microcracking during prolonged exposure to high temperatures



Thermosetting Matrix Materials

Some Details on Selected Thermosetting Matrix Materias Used in Aerospaced Composites

Matrix	Examples	Cure °C/kPa	Max (wet) Capability °C	Comment on Performance as Matrix
Epoxy	Hexcel 920	120/700	80+	Best properties all around
Epoxy	Hercukes 3501 Fiberite 934 Narmco 5208	180/700	100+	Excellent adhesion to fibers Easy to process, wide viscosity range, good wetting Excellent water and other chemical resistance Fairly low toughness, composites sensitive to impact damage Sensitive to UV exposure
Epoxy Toughened	Hercules 3502 Fiberite 977-2	180/700	100+	All of the above, plus: Improved tolerance to damage and increased moisture sensitivity
BMI	Hexcel F560	180/700 + postcure 200	230	Exceeds epoxy T capability Relatively easy to process Even less tough than epoxies Undergoes shrinkage during curing Prone to microcrack with thermal cycling
BMI Toughened	Cytec 5250-4	80/700	180	All of the characteristics listed for BMI, plus: More damage resistant Lower T capability than untoughened
Polyimide Condensation	PMR-15	300/400	320	Resistant to oxidation Similar properties to epoxy but overextended T range Low toughness and difficult to process

Thermosetting Matrix Materials

Properties of themosetting resins

Propiedad	Polyester	Phenolic	Epoxy	Bismaleimides	Polyimida
Density (g/cm ³)	1,1-1,5	1,3	1,1-1,4	1,2-1,3	1,2-1,9
Elastic Modulus (GPa)	1,3-4,5	4,4	2,1-6,0	3,4-4,1	3,0-3,1
Tensile Stregnth (MPa)	45-85	50-60	35-90	35-90	80-190
Fratcure toughness (MPa·m ^{-1/2})	0,5		0,6-1,0		
Fracture energy(kJ/m ²)			0,02		0,3-0,39
Coef.f thermal expansion(10 ⁻⁶ K ⁻¹)	100-200	45-110	55-110	60-65	14-90
Tg (°C)			120-190	275-300	300-350
Water Absortio (%)			2-6		4-4,5

Thermosetting Matrix Materials

Additives used to modify resin properties

- ✓ **Solvent** are added to reduce the viscosity before curing to aid in handling, wet-out, etc.
- ✓ **Plasticizers** are added to reduce the elastic modulus and increase the elongation to failure
- ✓ **Inert fillers**, including hollow spheres, are added to alter density, resin flow, cost, and effective modulus
- ✓ **Toughening agents** that precipitate from the reacting matrix during curing as fine particles, designed to modify the crack propagation properties in a cured matrix

Toughening mechanisms

- Random dispersion of a thermoplastic second phase in the initial formulation (polyethersulphone)
- Random dispersion of a rubber second phase *in-situ* during the polymerization process (carboxyl or amine terminated poly(butadiene-co-acrylonitrile:CTBN) rubbers.

Thermoplastic Matrix Materials

Currently a considerable effort to try to **substitute thermosetting matrices for thermoplastic matrices.**

The main advantages these resins are :

- Materials with unlimited shelf-life
- Storage at room temperature
- Better resistance to impact and moisture
- Toughness (strain at fracture 30% -100%)
- Good FST behavior (fire, smoke & toxicity)
- Processing speed
- reprocessability, recyclability

The main disadvantages of these resins are:

- High cost materials
- High processing costs (high temperature and pressure)
- Surface treatment of the carbon fibers
- Special Auxiliary Materials

Thermoplastic Matrix Materials

Unlike TS resins, TP are not cross-linked

Broad Classification: Amorphous and Semicrystalline

Amorphous TP

- High concentration of molecular entanglements
- Heating leads to disentanglement and change rigid solid → viscous liquid
- Low resistance to aggressive fluids
- Wide processing window
- Examples: PEI

Semicrystalline TP

- High degree of molecular order and alignment
- May have anisotropic properties (crystalline lamellar units oriented)
- Very good resistance to aggressive fluids
- The processing window can be critical (especially cooling rate)
- Examples: PEEK, PPS

Thermoplastic processed by injection molding:
Reinforced with carbon fiber or glass:
Examples: PEEK, PA, PTFE

Polyether ether ketone (PEEK)

Polyphenylene sulfide (PPS)

Polysulfone PSF

Poethersulfones PES

Polyimide PI

Polyethylenimine (PEI)

Thermoplastic Matrix Materials

Some Details on Selected Thermoplastic Matrix Materias used in Aerospace Composites

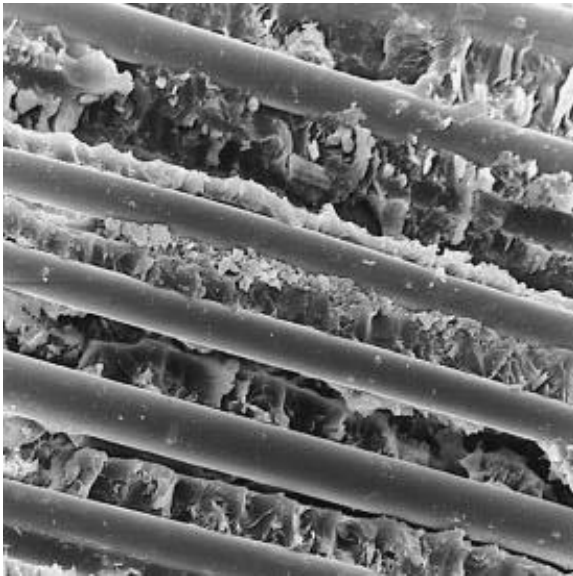
Polymer	Example	Process T °C	T _g °C	Comment on Performance as Matrix
PEEK	Victrex	400	145	Excellent mechanical properties, including toughness. For application until 120 °C. Highly resistant to damage. Excellent fire resistance. Most widely used for high-performance composites
PPS	Ryton	340	90	Good strength, stiffness and T capability. Some grade have low viscosity. Resistant to most fluids, but attacked by some solvents. Good fire resistance. Low impact resistance
PSF	Udel	400	190	
PES	Vitrex	400	230	
PI	Kapton	390	320	Selected for highest T applications. Highly viscous. Difficult to process. Select for high-T applications.
PEI	Ultem	370	215	

The interface region

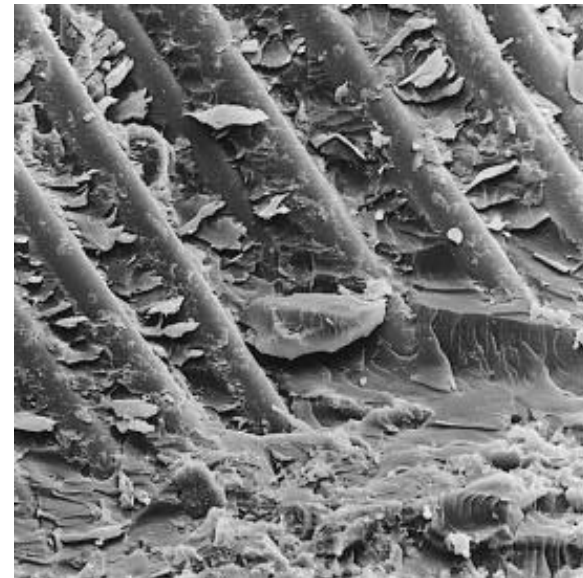
INTERFACE: The planar region that separates two components.

Interphase, a new concept, a region of **space** characterised by a volumen and property gradients, that separates two components

The final properties of a composite with a polymeric matrix reinforced with fibers depends upon the interfacial adherence



Poor adherence matrix-fibers



Good adherence matrix-fibers

The interface region

Bonding mechanism

In most treatment, is assumed than the interfacial bond is “perfect”. This means that there is **no *debonding, cracking or sliding***

In practice:

!! many important phenomena may take place at the interface !!



PLASTIC DEFORMATION OF THE MATRIX: CAN INFLUENCE ON FAILURE

A bonding system between the matrix and the fiber is necessary to consistently achieve the ideal mechanical properties

To Formation of Interfacial bonds:

- Adsorption and wetting
- Interdiffusion and chemical reaction
- Electrostatic attraction
- Mechanical keying

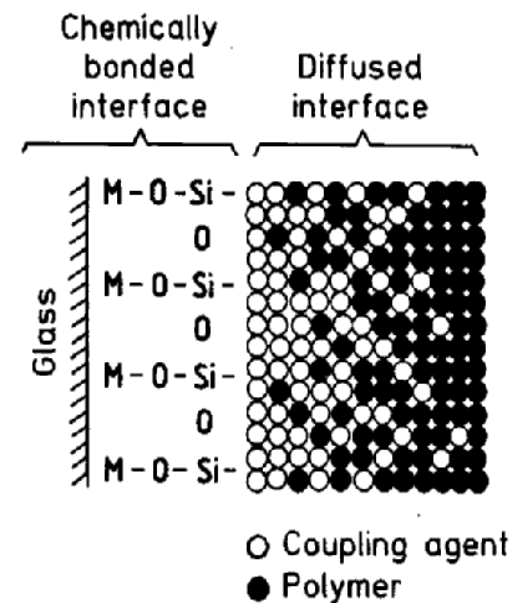
The interface region

CONTROL OF BOND STRENGTH: Coupling agents, toughness-reducing coating and interfacial chemical reaction and diffusion barrier coating

Glass fibres: surface sizing

Strength of glass fibres is determined by: **damage** during processing, and the **adsorption of water** over their surface

- **Role of sizings**
 - **binds** the filaments ($d = 10\text{-}20\text{ mm}$) together for ease processing,
 - **lubricates** fibres so they can withstand abrasion,
 - imparts **anti-static** properties
 - provides some **chemical functionalities** to the surface to improve interfacial bond strength.
- Sizing normally contains a **film-forming polymer**, such as polyvinyl acetate, and a **coupling agent** (organosilane, titanate or chromate).



Mean elastic properties of composites

Composite material properties:
mean property of its individual components:

Mean properties depend upon the microstructural geometry:

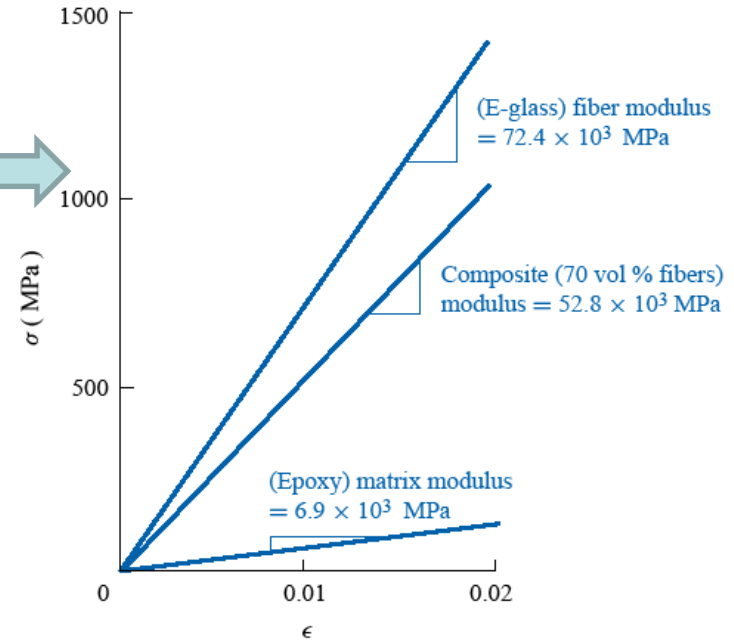
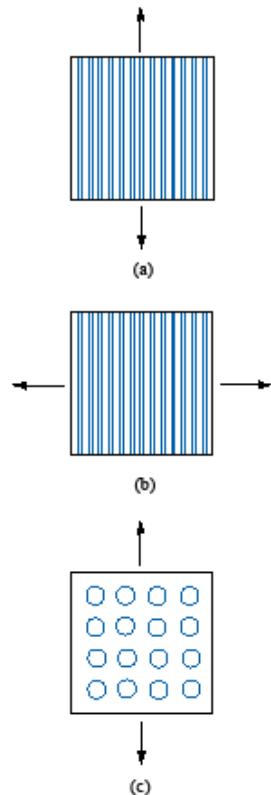


FIGURE 10-11

Three ideal geometries:

- a) **Fiber-reinforce composite:** Direction parallel to continuous fibers
- b) **Fiber-reinforce composite:** Direction perpendicular to continuous fibers
- c) **Particle Reinforced composite** (uniformly dispersed phase)

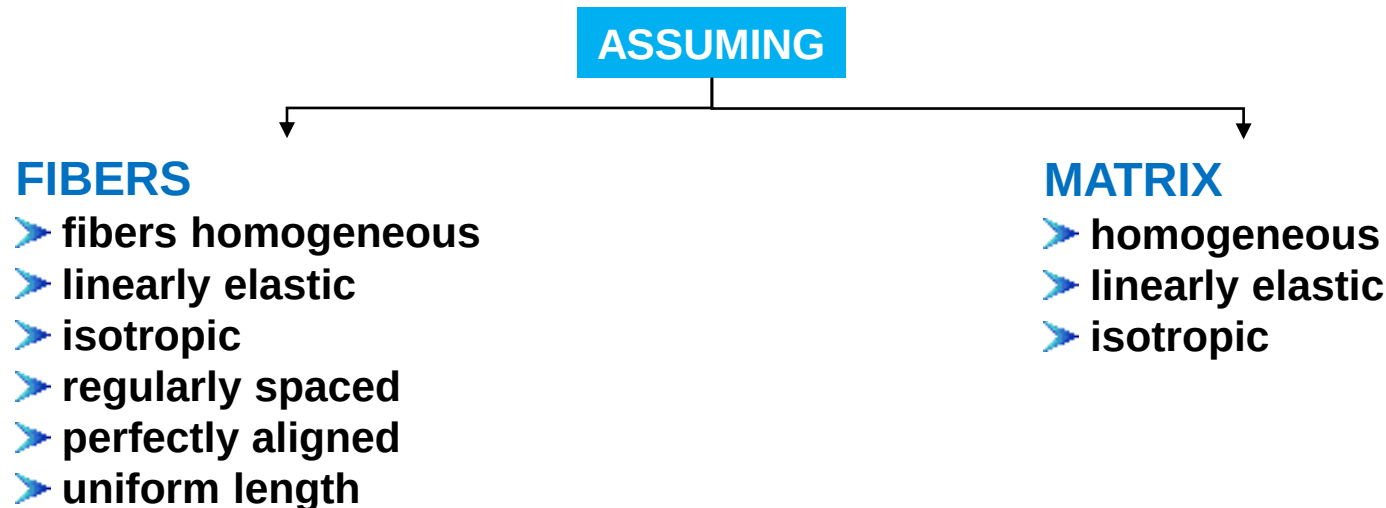
Mean elastic properties of composites

Micro analyses studies for fiber composites

MICROMECHANICS : Utilizes microscopic models of composites, in which the fibers and the matrix are separately modelled.

Micromechanics can be approach in three ways:

1. The **mechanics of materials**: predict the behaviour of simplified models of the composite material
2. The **theory of elasticity**: predict upper and lower bound exact analytical or numerical solutions
3. The **finite-element**: based on 2-dimensional or 3-dimesional models of varing degrees of sophistication



Mean elastic properties of composites

COMPOSITE MATERIALS REINFORCED WITH FIBERS

ISO STRAIN conditions

Direction parallel to continuous fibers (1-direction):

→ The axial strain:

Assuming good bond:

$$\varepsilon_C = \varepsilon_f = \varepsilon_m$$

$$F_C = F_f + F_m \quad \text{such that} \quad \sigma = \frac{F}{A} \Rightarrow \sigma_C A_C = \sigma_f A_f + \sigma_m A_m$$

$$\sigma_C V_C / l_c = \sigma_f V_f / l_f + \sigma_m V_m / l_m \Rightarrow \sigma_C V_C = \sigma_f \cdot V_f + \sigma_m \cdot V_m \Rightarrow \sigma_C \frac{V_C}{V_C} = \sigma_f \cdot \frac{V_f}{V_C} + \sigma_m \cdot \frac{V_m}{V_C}$$

Such that $V = A \cdot L$ and $L_C = L_f = L_m$ A can be replaced by volume fraction V

where σ_C , σ_f , and σ_m are the stresses and A_C , A_f and A_m are the areas of the CM, fiber, and matrix, respectively

→ The overall stress

$$\sigma_C = \sigma_f V_f + \sigma_m V_m$$

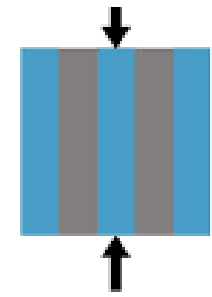
since $V_C = 1$

$$\sigma_{1C} = (1 - V_f) \sigma_{1m} + V_f \sigma_{1f}$$

* sub index 1 : direction parallel to fibers



Isostrain

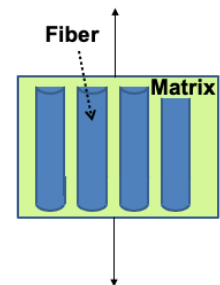


Isostrain

$$\varepsilon = \varepsilon_{\text{scaffold}} = \varepsilon_{\text{de novo bone}}$$

$$\sigma = \sigma_{\text{scaffold}} + \sigma_{\text{de novo bone}}$$

Displacement continuity across all layers



Elastic deformation of long-fibre composites

COMPOSITE MATERIALS REINFORCED WITH FIBERS

ISO STRAIN conditions

$$\frac{\sigma_C}{\varepsilon_C} = \frac{\sigma_f V_f}{\varepsilon_f} + \frac{\sigma_m V_m}{\varepsilon_m} \Rightarrow E_{C1} = \frac{\sigma_{1C}}{\varepsilon_{1C}} = \frac{[(1 - V_f)\sigma_{1m} + V_f\sigma_{1f}]}{[\sigma_{1f}/E_f]} = \left[\frac{(1 - V_f)\sigma_{1m}}{\sigma_{1f}} + V_f \right]$$



Using the ratio between the stresses in the components:

→ The **Young's modulus** of the composite (direction 1)

$$E_C = E_f v_f + E_m v_m$$

"rule of mixtures"

$$E_{1C} = (1 - V_f)E_m + V_f E_f \quad (\text{eq 1})$$

For a composite in which the **fibres are much stiffer** than the matrix ($E_f \gg E_m$), the reinforcement is subject to much higher stresses ($\sigma_{1f} \gg \sigma_{1m}$) than the matrix and there is a redistribution of the load

for typical fiber composites with $V_f=50-60\%$, the matrix has only a small effect upon $E_1 \Rightarrow E_1 \approx E_f V_f$



$$E_f \gg E_m \rightarrow E_c \approx V_f E_f$$

* sub index 1 : direction parallel to fibers, Longitudinal Modulus E_1

Elastic deformation of long-fibre composites

COMPOSITE MATERIALS REINFORCED WITH FIBERS

ISO STRESS conditions

Direction perpendicular to continuous fibers (2-direction)

$$\sigma_C = \sigma_f = \sigma_m$$

$\sigma_{2C} = \sigma_f = \sigma_m$ and assuming that cross-sectional area under does not change after the stress is applied $\Rightarrow$

the length of each component:

$$L_m = A_m L_C$$

$$L_f = A_f L_C$$

$$\text{and } \Delta L_C = \Delta L_m + \Delta L_f$$

$$\frac{\Delta L_C}{L_C} = \frac{A_m \Delta L_m}{L_m} + \frac{A_f \Delta L_f}{L_f}$$

Given that A can be replaced by volume fraction V

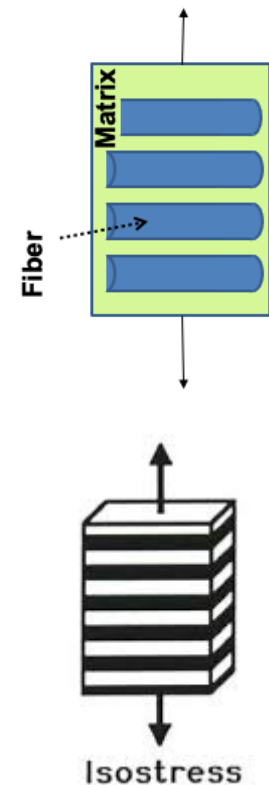
→ the **overall net strain** is

$$\boxed{\varepsilon_C = \varepsilon_f V_f + \varepsilon_m V_m} \Rightarrow \varepsilon_{2C} = V_f \varepsilon_{2f} + (1 - V_f) \varepsilon_{2m}$$

and the transverse strains for the fiber, matrix, and composite are:

$$\varepsilon_f = \frac{\sigma_2}{E'_f}, \quad \varepsilon_m = \frac{\sigma_2}{E_m}, \quad \varepsilon_2 = \frac{\sigma_2}{E_2}$$

* sub index 2 : direction perpendicular to fibers



Elastic deformation of long-fibre composites

COMPOSITE MATERIALS REINFORCED WITH FIBERS

ISO STRESS conditions

When a **stress** applied in the 2-direction (isostress)

$$\sigma_{2C} = \sigma_{2f} = \varepsilon_{2f} E_f = \sigma_{2m} = \varepsilon_{2m} E_m$$

$$\frac{\sigma}{E_C} = \frac{\sigma v_f}{E_f} + \frac{\sigma v_m}{E_m}$$

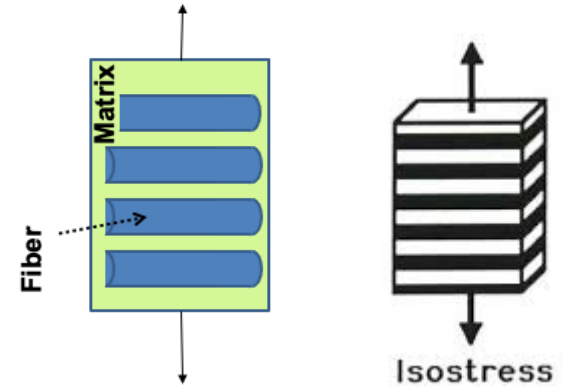
Dividing by σ and rearranging the terms

$$\frac{1}{E_{2C}} = \frac{V_f}{E_f} + \frac{V_m}{E_m} \quad (\text{eq 2})$$

→ the **composite modulus**:

$$E_C = \frac{E_f E_m}{v_f E_m + v_m E_f}$$

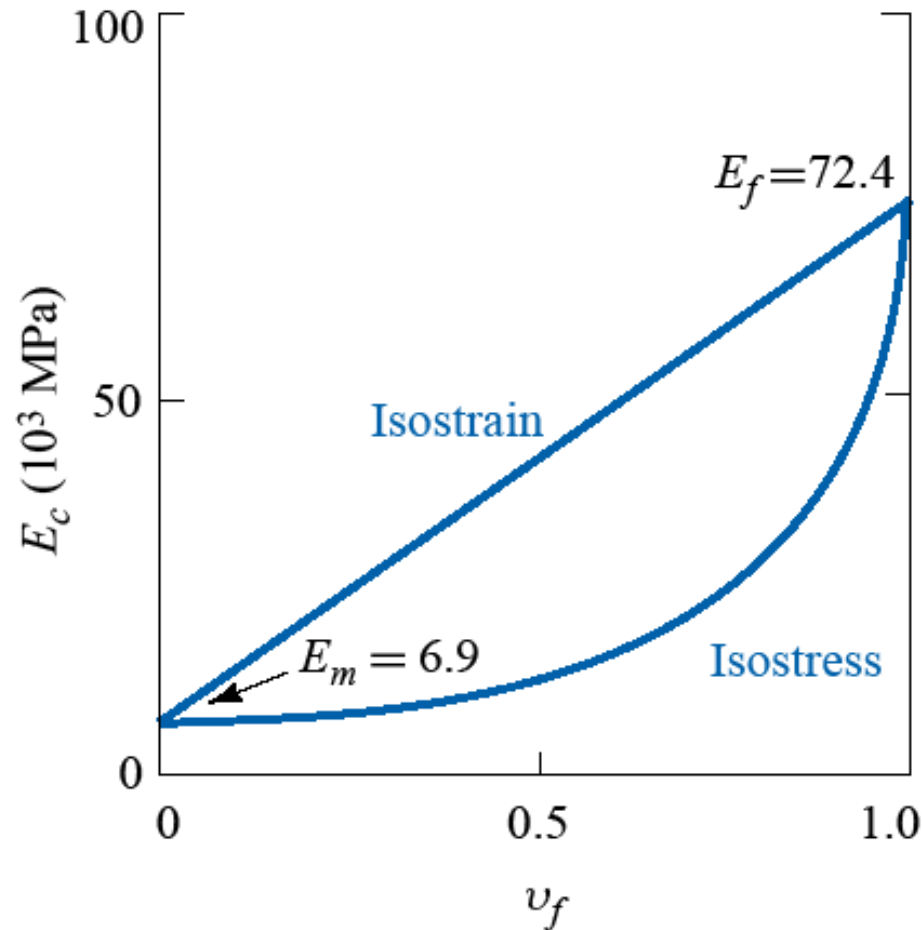
for typical fiber composites with $V_f=50-60\%$: $E_2 \approx \frac{E_m}{V_m}$



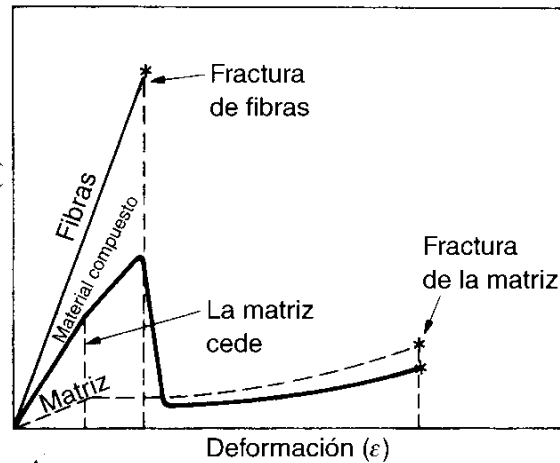
* sub index 2 : direction perpendicular to fibers, Transverse Modulus E_2

Elastic deformation of long-fibre composites

Example: fiberglass reinforced epoxy resin



Elastic deformation of long-fibre composites



Let there be a composite with modulus E_C and strength σ_C . The matrix has E_m , σ_m and $\sigma_{y,m}$ and the fiber E_f and σ_f .

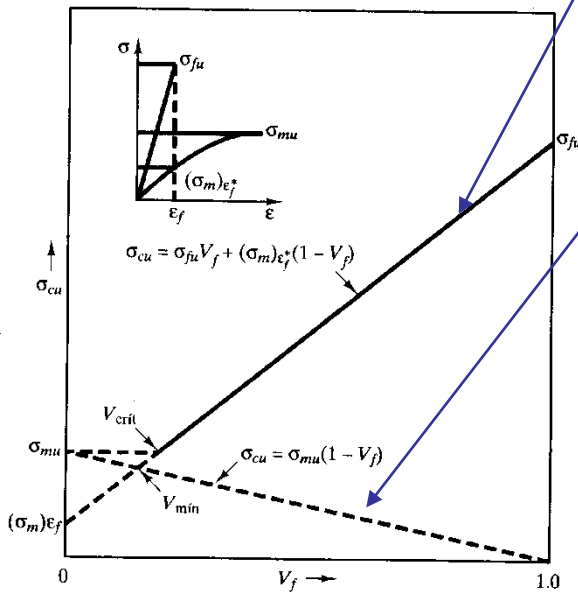
Strength

For similar reasons as those explained before, the strength σ_C of an unidirectional composite follows the same rule of mixtures

$$\sigma_C = \sigma_f V_f + \sigma_{y,m} (1 - V_f)$$

Fibers break before the matrix because they are more brittle. When this happens, the equation transforms to:

$$\sigma_C = \sigma_m (1 - V_f)$$



A minimum V_f (critical) is required. If V_f is lower than the critical value, the strength of the materials is reduced.

Mean elastic properties of composites

PARTICLE-REINFORCED COMPOSITE MATERIALS

RULE OF MIXTURES: The properties of a composite material depend upon the relative quantities and properties of its constituents

$$P_{CM} = \sum f_i p_i$$

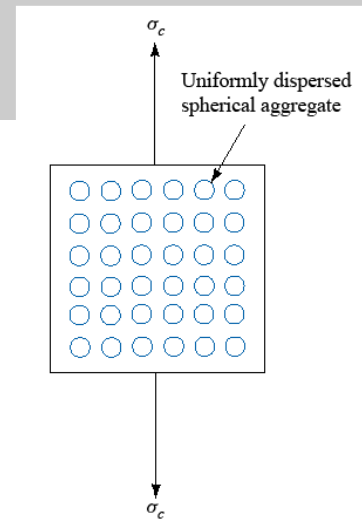
P_{CM} = property of the CM
 f = volume fraction of each i ($\sum f_i = 1$)
 i = i -th component

CM with homogeneously distributed particles → **Isotropic** properties

A thorough treatment of this system is complex and depends on the nature of the dispersed phase and the matrix. An approximation is given by the equation:

$$E_C^n = v_l E_l^n + v_h E_h^n$$

l = lower modulus phase
 h = higher modulus phase



Mean elastic properties of composites

PARTICLE-REINFORCED COMPOSITE MATERIALS

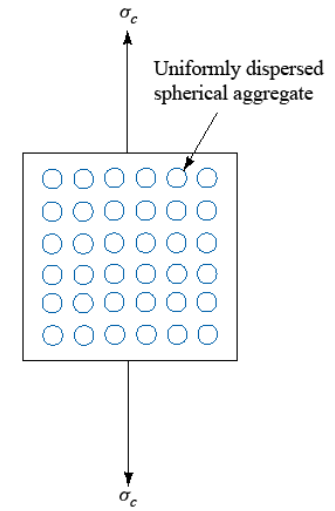
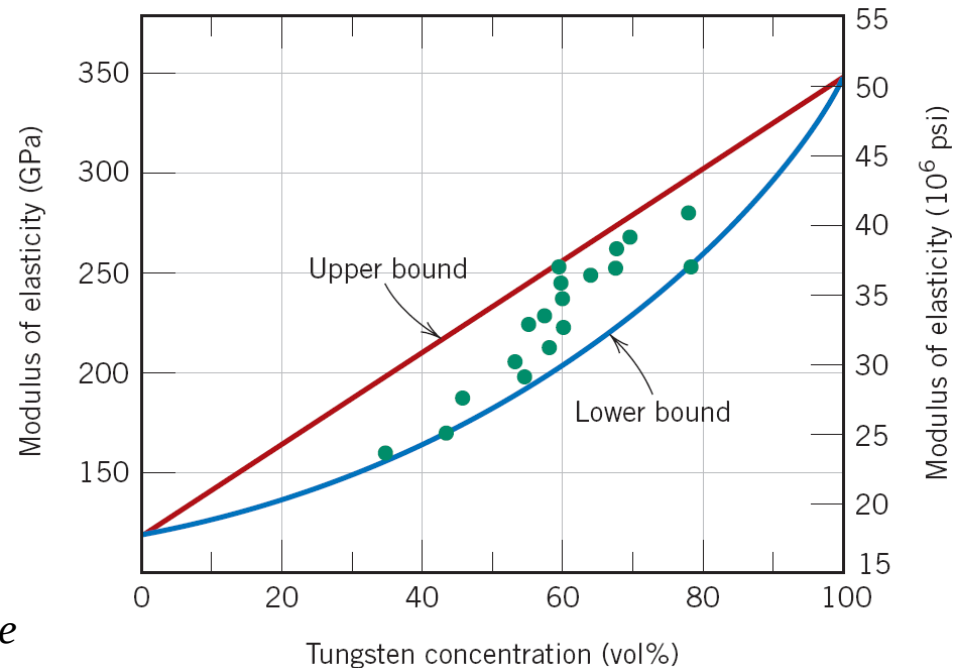
E value lies between two limits:

Upper bound (máximo) : $n = 1$ $E_C = E_l v_l + E_h v_h$

Lower bound (mínimo) : $n = -1$ $E_C = \frac{E_h E_l}{v_h E_l + v_l E_h}$

Upper bound: $n = 1$

Lower bound: $n = -1$



$l =$ lower modulus phase
 $h =$ higher modulus phase

Improvement of mechanical properties

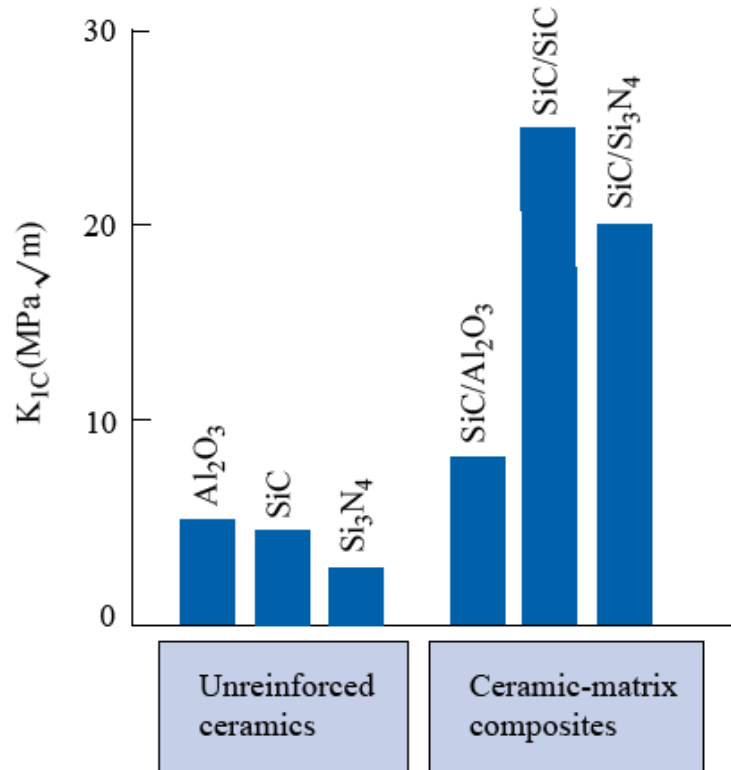


FIGURE 10-18

Increase of toughness to fracture of some ceramic matrix composites

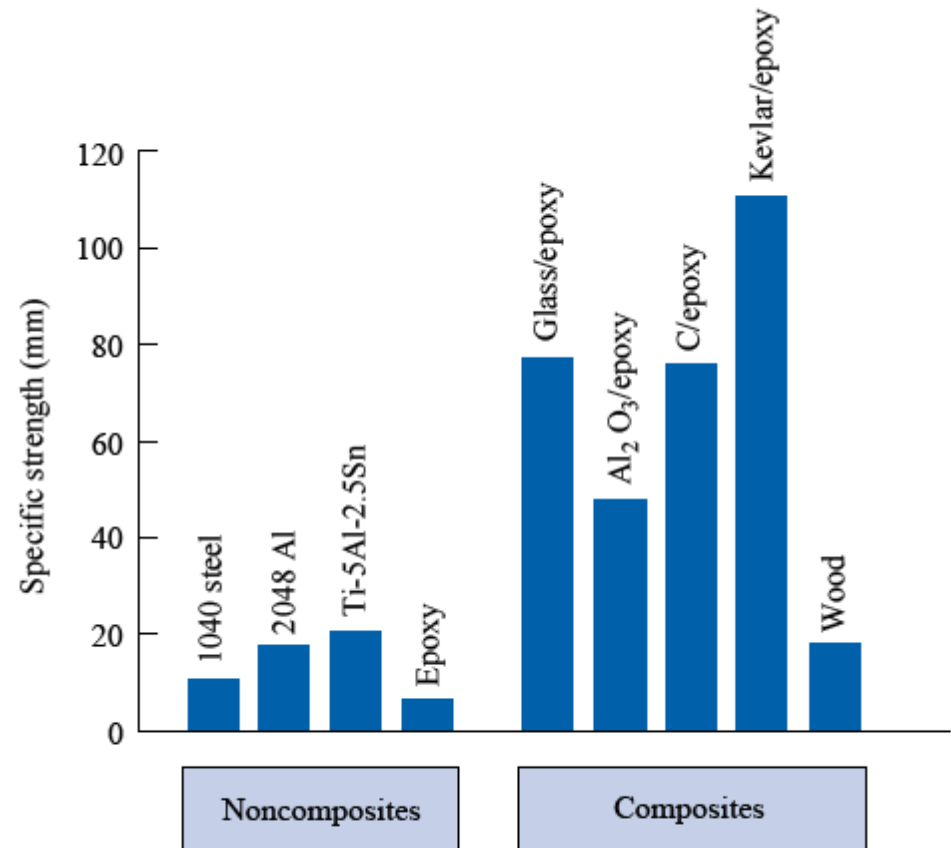


FIGURE 10-19

Increase of strength of epoxy resin with different reinforcements